

Progress Report

YNLT aquatic plant classification

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Finished

Request	Status
Reframe contribution (small data + lab-to-field + YNLT)	Done in title, abstract, intro
Shorter intro + related methods	Done
YNLT math (confidence, margin, entropy, trigger)	Done
Ablation: backbone / gated head / YNLT	14 runs finished
Stronger baselines (ResNet, EfficientNet, ViT, ConvNeXt)	Trained
Repeated splits, mean $\pm$ std (seeds 8–12)	Trained
Computational cost / robot claim	Measured; robot claim removed
Check references	11 verified, 11 still open
Read 2 papers / week	Swin + DINOv2 done; BioCLIP unfinished

Cost measured on RTX 3070 Ti: **4.47 G MACs**, **5.0 ms** per 224 pass (draft said ~ 322 M). One reviewed image ≈ 187 ms.

New results (not yet in the paper)

Trained on **203 lab** / **42 field** images. Paper still uses **230** / **33**. These numbers do not replace Table 1 until we confirm the official set.

Model (ConvNeXt-V2, 5 seeds)	Lab %	OOD %
Linear	65.9 ± 29.4	20.0 ± 12.3
Linear + YNLT	69.3 ± 30.1	18.1 ± 17.0
Gated attention	80.0 ± 6.8	19.1 ± 7.5
Gated + YNLT	82.0 ± 6.4	18.1 ± 8.5

Still to do

- Put the new tables into the paper.
- Confirm which image list is official (230/33 vs 203/42).
- Recover Paper/**figs/** and compile the draft.
- Fix the 11 unverified citations.
- Finish BioCLIP reading notes.

Questions

- Which dataset should the paper use?
- Can I have Aaryan's original `ref.bib`?